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New palaeomagnetic data from Argun terrane. Testing its association with Amuria and the Mongol–Okhotsk Ocean

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SUMMARY

In this study, we present new palaeomagnetic and geological data obtained from Ediacaran and Cambrian sedimentary rocks of Argun terrane, which is traditionally considered a key element of the hypothetical Amuria composite continent. Since 1990, when Amuria was first proposed in palaeogeographic reconstructions, it became one of the principle members in the global palaeotectonic schemes. A scenario when collision of Amuria with Siberian margin resulted in formation of the Mongol–Okhotsk Ocean is universally accepted and supported by majority of researchers. However, time of Amuria's final assembly and relative position of the blocks within Amuria before the collision with Siberia is still a topic of debate. Questions about principal allocation of Argun terrane and its relation to Amuria during the late Neoroterozoic–Cambrian are addressed in this study. Palaeomagnetic poles for the Ediacaran–early Cambrian rocks of Argun terrane differ within an error from coeval poles from Siberia indicating that Argun terrane could have been located similar to its present-day position with respect to Siberia already at 560–525 Ma. This observation calls into question association of Argun terrane with Amuria, which in classic reconstructions is usually placed close to the North China Craton. It also questions our current understanding of the Amuria palaeocontinent and consequently, accuracy of global palaeogeographic reconstructions for the late Neoproterozoic–Cambrian in general, and those of the eastern part of the Central Asia in particular.

Key words: Palaeomagnetism; Continental margins: covergent; Continental margins: divergent; Continental tectonics; Cratons.

1 INTRODUCTION

The majority of palaeogeographic reconstructions of the eastern part of the Central Asia, denoted as the Mongol–Okhotsk Fold Belt, (MOFB) presume an involvement of the large Amuria continent. This supercontinent, first proposed by Zonenshain *et al.*(1990) included Argun, Central-Mongolian, Jiamusi-Burea, and associated terranes that constituted the carcass of the present-day eastern part of the Central Asia. Classical reconstructions assume that the basement of these terranes is mainly composed of Archean-Palaeoproterozoic complexes that assembled during the late Neoproterozoic–early Cambrian. Based on the widespread of Neoproterozoic–Cambrian carbonate sedimentary rocks, it was proposed that Amuria could have been a part of the North China Craton or located close to it far away from Siberia (Zonenshain *et al.* 1990). In the late Palaeoproterozoic palaeogeographic reconstructions Central-Mongolian, Jiamusi-Burea and Argun terranes are interpreted as independent microcontinents, which along with Kazakh superterrane were located within the Palaeo-Asian Ocean between continental margins of Siberia and North China. Gradual convergence of the microcontinents with Siberia and consequent collision with island arcs of Southern Mongolia resulted in formation of a large accretional belt that included Amuria. This accretional structure as a huge peninsula stretched from the modern SW Siberian margin far into the ocean separated a large bay, known as the Mongol–Okhotsk Ocean. Closure of the Mongol–Okhotsk Ocean, caused by convergence of Siberia and Amuria, which probably have started in the Devonian (Donskaya *et al.* 2013), resulted in formation of a narrow strip of accretion–collisional structures of the MOFB and the Mongol–Okhotsk suture in the Jurassic–early Cretaceous (Zonenshain *et al.* 1990; Parfenov *et al.* 1999; Kravchinsky *et al.* 2002; Metelkin *et al.* 2010; Wilhem *et al.* 2012).

Since it was first proposed in palaeogeographic reconstructions (Zonenshain *et al.* 1990), Amuria played a key role in global

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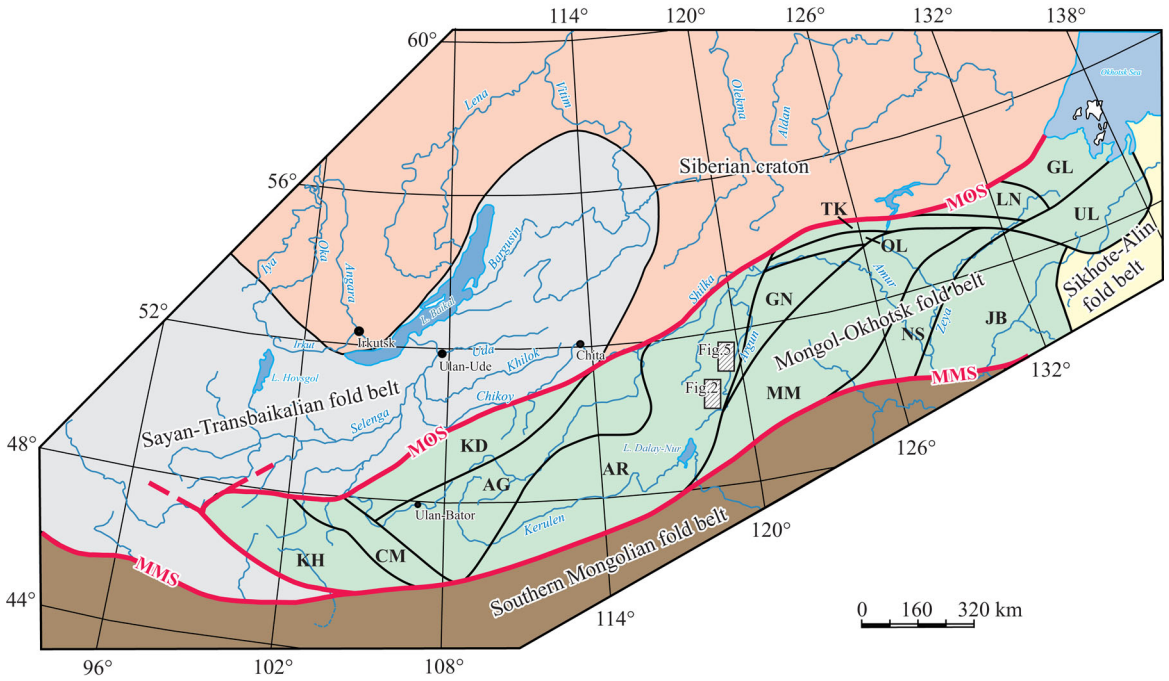


Figure 1. A simplified map of the main terranes of the MOFB and adjacent structures. Abbreviation: KH—Khangay terrane, CM—Central Mongolian terrane, KD—Khentey–Daur terrane, AG—Agin terrane, AR—Argun terrane, OL—Oldoy terrane, GN—Gonzha terrane, MM—Mamyn terrane, NS—Nora–Sukhotin terrane, JB—Jiamusi–Bureya terrane, TK—Tukuringra terrane, LN—Lana terrane, GL—Galam terrane, UL—Ulban terrane, MOS—Mongol–Okhotsk suture and MMS—Main Mongolian suture

palaeotectonic reconstructions (Kravchinsky *et al.* 2002; Golonka *et al.* 2006; Metelkin *et al.* 2010). However, time of Amuria's final assembly and relative position of the blocks within Amuria before the collision with Siberia is ambiguous and its assessment requires solid evidence based on geology and reliable palaeomagnetic data. There are some problem in understanding of formation mechanisms of numerous orogenic systems that include terranes of different origin such as Central-Mongolian, Argun, Jiamusi-Bureya and others (Yarmolyuk *et al.* 2014). For example, in spite of traditional view that early Precambrian crystalline complexes form the cores of these terranes, it has been recently shown that protoliths of metamorphic rocks have formed during the Neoproterozoic. In addition, some overlying Neoproterozoic–Palaeozoic sedimentary rocks have been deposited within common with Siberia basins. (Wilde *et al.* 2003; Kotov *et al.* 2009, 2013; Khanchuk *et al.* 2010; Sorokin *et al.* 2010; Metelkin *et al.* 2013). These studies call into question the traditional view on the existence of Palaeoproterozoic crystalline basement, and consequently, the origin of the main parts of Amuria, as well as the validity of allotment of such composite terrane in principle. Palaeonagnetism plays a key role in understanding the internal structure, palaeogeography and the evolution of Amuria. However, available data set is scarce and consists of a very limited number of determinations for only a few stratigraphic units within the MOFB. The shortage of palaeomagnetic data is especially notable for the Neoproterozoic–Palaeozoic period of the MOFB geological history (Bretshtein & Klimova 2007; Metelkin *et al.* 2013).

In this paper, we present the results of a comprehensive geological and palaeomagnetic study of various rocks from Argun terrane, which is considered a key element of the hypothetical Amuria. Preliminary palaeomagnetic results have been published in Metelkin *et al.* (2013). Here, we report an extended palaeomagnetic data set and more detailed analyses that include new palaeomagnetic pole positions for the late Ediacaran and early Cambrian. Based on this data, we test association of Argun terrane with Amuria and their mutual position within the Palaeo-Asian and Mongol–Okhotsk oceans.

2 GEOLOGICAL BACKGROUND

2.1 General overview

To date, there are several tectonic models of the MOFB that differ in details of interior structure as well as in outer boundaries of the belt (Zonenshain *et al.* 1990, Sengör *et al.* 1993; Gordienko 1994, 1996; Zorin *et al.* 1998; Parfenov *et al.* 1999, 2003). In our interpretation, tectonic elements of the orogenic belt are in contact with early Precambrian crust of Siberian craton and with Neoproterozoic to late Palaeozoic Sayan–Transbaikalian assembly of terranes. The contact is manifested along the Mongol–Okhotsk suture at the north and northwest (Fig. 1). The interior structure of the belt is built of several terranes of different ages and origin, involved in the closure of the Mongol–Okhotsk Ocean. To the south, these terranes are welded through the Main Mongolian suture to the early and late Palaeozoic orogenic structures of the Central, Southern and Inner Mongolias that separate the belt from the North-China craton.

Argun terrane, located in Transbaikalia and adjacent regions of Mongolia, occupies the central place within the MOFB (Fig. 1). To the west, it is bounded by Ordovician and Devonian island arc complexes of Agin terrane (Gordienko 2006; Lykhin *et al.* 2007; Gordienko *et al.* 2012). To the north, in the lower reaches of the Shilka River, Argun terrane is almost directly connected to metamorphic complexes of the Siberian craton. A narrow strip of mylonites that correlate with Mongol–Okhotsk suture has been recognized there. Eastern margin of Argun terrane consists of Tukuringra and Oldoy terranes that are considered fragments of the Palaeozoic–Mesozoic active margin, and Mamyn metamorphic promontories,

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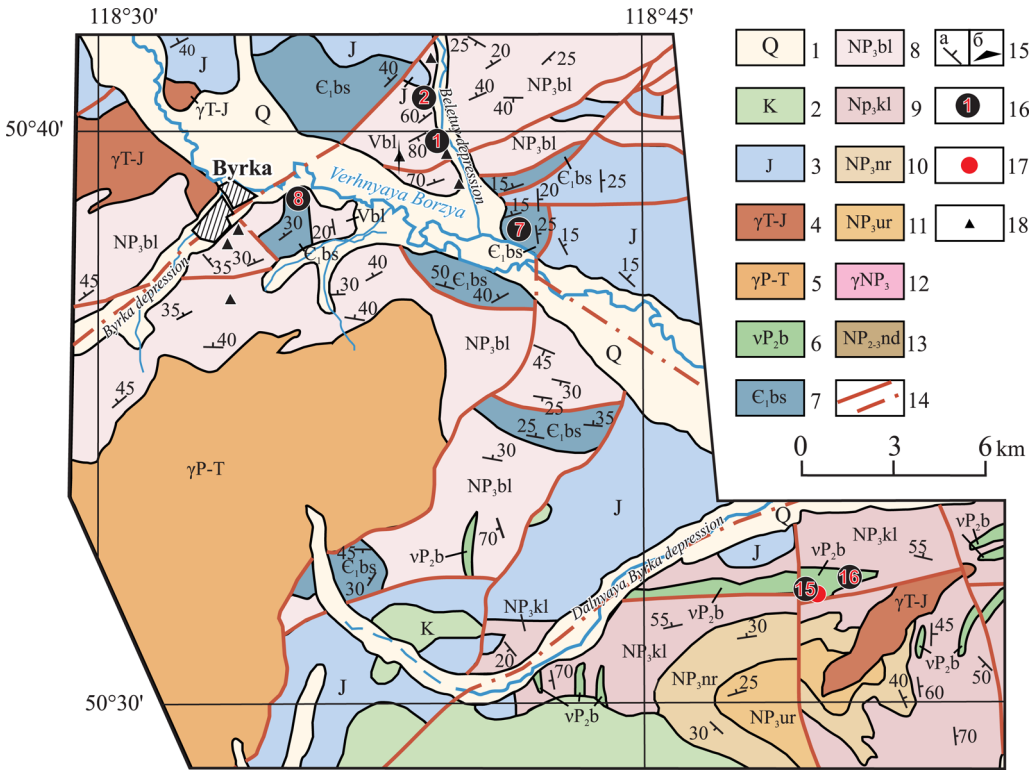


Figure 2. Geological map of Byrka area adopted from maps of Chita Geologic Survey (Chitageols'emka). Legend: 1—quaternary deposits (Q); 2—cretaceous rocks (K); 3—Jurassic rocks (J); 4—Shakhtaminsk granites (γT-J); 5—Undin granites (γP-T); 6—Byrka dolerites (vP_2b); 7—Bystrin carbonate-terrigenous formation ($Є_1bs$); 8—Beletuy carbonate-terrigenous formation (NP_3bl); 9—Klichka carbonate-terrigenous formation (NP_3kl); 10—Nortui carbonate-terrigenous formation (NP_3nr); 11—Urulyunguy shale-terrigenous formation (NP_3ur); 12—Urtui granites (γNP_3); 13—Nadarov gneiss-schist-dolomite formation ($NP_{2\text{-}3}nd$); 14—faults; 15—bedding: (a) structural tilt and (b) foliation and gneissosity; 16—palaeomagnetic sampling sites, number inside according to Table 2, 17—isotope-geochemistry and geochronology sampling sites and 18—fauna fossils sampling sites.

reworked during the Palaeozoic and Mesozoic tectonic and magmatic events (Khanchuk *et al.* 2006).

Russian part of Argun terrane is composed of a variety of Precambrian to Mesozoic sedimentary, sedimentary-volcanic, volcanic, intrusive and metamorphic rocks.

Mesozoic complex is characterized by voluminous early and middle Jurassic folded flysh and molasses that unconformably overlay the oldest Neoproterozoic–early Palaeozoic basement. Rift-related depressions are filled with Jurassic and early Cretaceous volcanic rocks (Metelkin *et al.* 2010). Intraplate granitic bodies are also common.

Palaeozoic complex is characterized by abundance of Silurian to Devonian volcanic rocks of oceanic and island arc type. Formerly, these rocks have been interpreted as Neoproterozoic in age. However, Sm–Nd analysis gave the protolith T_{Nd}(DM) model age of 583 Ma. Positive values of T_{Nd} ranging from +9.2 to +10.6 suggest an association of mantle material with subduction processes. U–Pb geochronological studies of zircons resulted in the absolute age of the complexes in a range of 390–420 Ma (Dril & Golubev 2003; Ruzhentsev & Nekrasov 2009).

In addition to volcanic series, thick Silurian, Devonian and early Carboniferous carbonate–terrigenous sections were recognized in geological composition of the framing complex. Similar sequences were identified on the territory of Oldoy terrane. According to detrital zircon isotope age study (Sorokin *et al.* 2015), these rocks were being deposited from middle Devonian in active continental margin environment.

Neoproterozoic metamorphic rocks, intruded by coeval granites, constitute the major and oldest part of Argun terrane. The majority of metamorphic rocks were previously considered early Precambrian protrusions of crystalline basement of Argun microcontinent (Zonenshain *et al.* 1990). However, recent isotope-geochemistry and geochronology data indicate Mesoproterozoic and Neoproterozoic age of the protolith and granites, respectively (Golubev *et al.* 2010).

Finally, the Ediacaran–Cambrian complex can be considered an independent one. It is characterized by abundance of terrigenous–carbonate rocks. Traditionally, the complex is associated with a shallow-sea sedimentary basin isolated from the Siberian continental margin (Zonenshain *et al.* 1990). Particularly, it is thought that the basin could have been a part of continental margin basins of Amuria, located on the opposite from Siberia side of the ocean.

In order to facilitate our understanding of tectonic evolution and palaeogeography of Argun terrane, we carried out a detailed geological, geochronology and palaeomagnetic study of Ediacaran to Cambrian rocks of the Beletuy and Bystrin formations exposed in Byrka and Zerentuy areas of the Russian part of Argun terrane (Fig. 1).

2.2 Byrka area

In this geologically complex region, the Bystrin formation crops out fragmentarily, while the Beletuy formation is well exposed and occupies considerable part of the area (Fig. 2). The Beletuy formation conformably overlies older Precambrian, variably metamorphosed terrigenous sedimentary rocks and crops out along both banks of the Verhnyaya Borzya River near the Byrka village (Figs 2 and 3).

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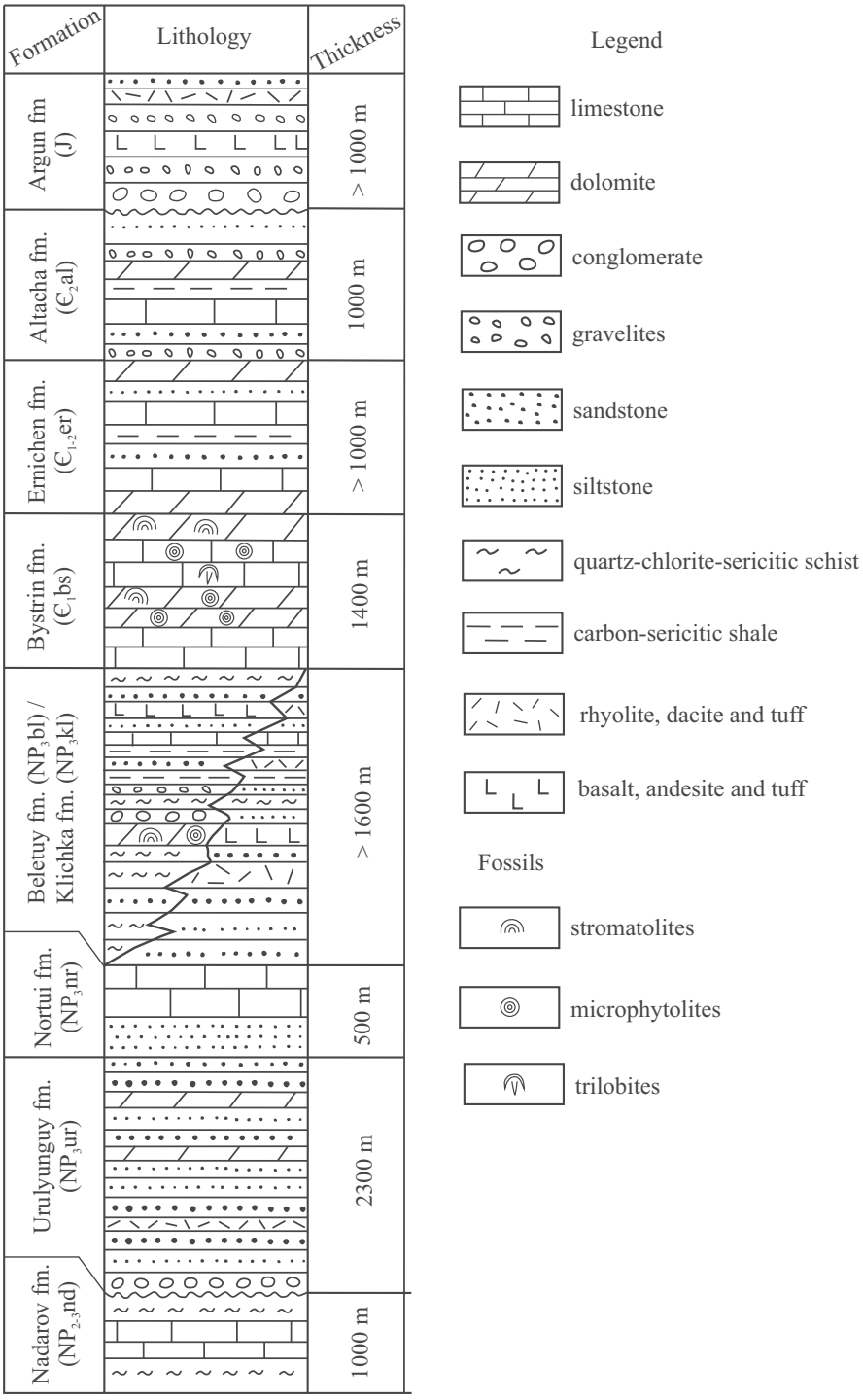


Figure 3. A scematic lithostratigraphic chart for the Byrka and Zerentuy areas.

The Beletuy formation consists of a wide variety of rocks that also underwent some metamorphism that did not exceed the green schist facies. Shales with interbedded sandstones and green-grey siltstones constitute the base of the formation. The middle part of the formation is characterized by abundance of dolomites and limestones. Stratigraphically higher, quarts and arkose gravelites and sandstones of variable grain sizes, sometimes interbedded with shales are present. Individual, relatively small (up to a few metres) lenses of intraformational conglomerates were defined. Comparatively thin (100–200 m) piles of schistose basalts are common for the upper part of the Beletuy formation. Similar rocks were identified on the territory of the adjacent Mamyn terrane. U–Pb dating of zircons from Mamyn volcanics resulted in the absolute age of 546 ± 14 Ma (Sorokin & Kudryashov 2015).

At the southeast of the Byrka area, the Beletuy formation is replaced by its facial analogue—the Klichka formation (Figs 2 and 3), which is mainly composed of sandstones and siltstones with interbedded schistose basalts, andesites, rhyolites and tuffs. Doleritic sills and thin dykes, previously interpreted as comagmatic with the basalts of the Beletuy and Klichka formations, are

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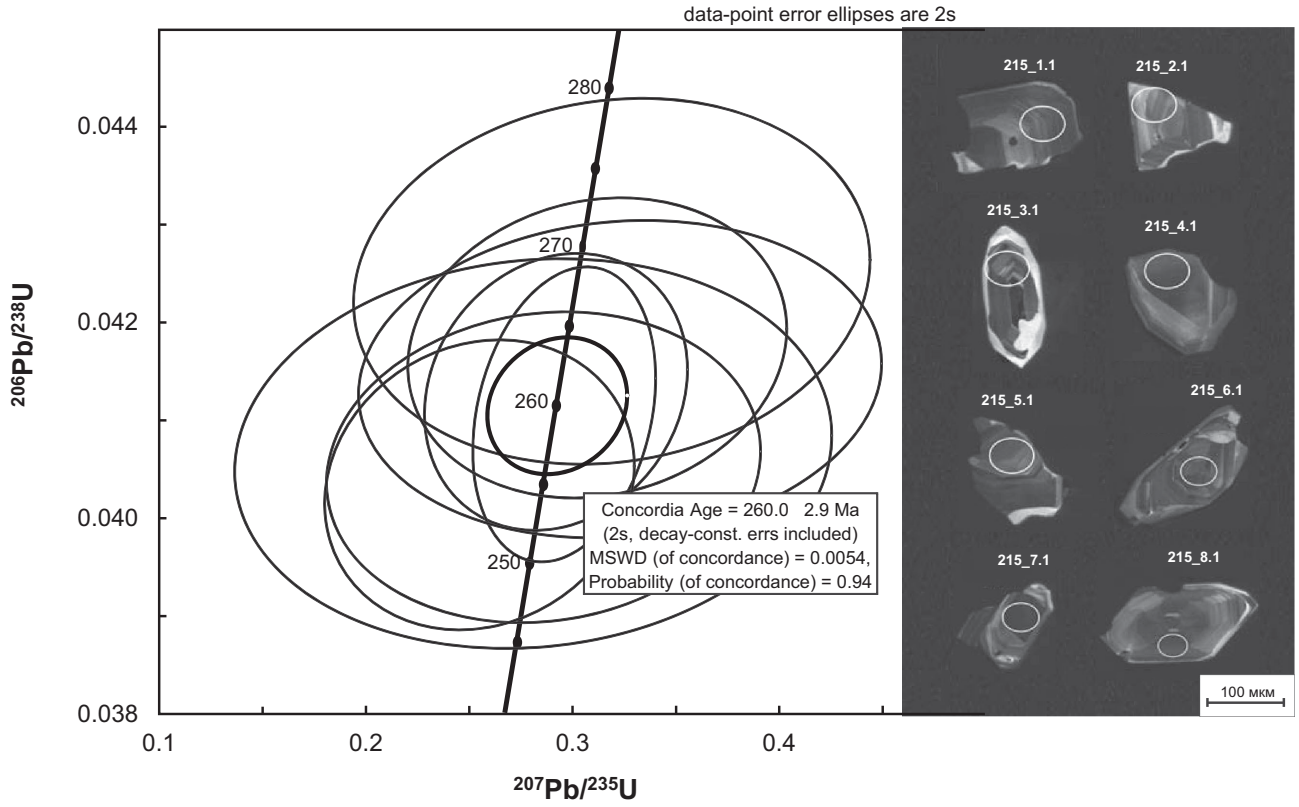


Figure 4. Zircon U–Pb concordia diagram for Byrka dolerite samples. Sampling site see on Fig. 2.

also common. However, our U–Pb study of zircons from dolerites indicates their late Permian age of 260 ± 2.9 Ma (Fig. 4). We interpret these rocks as a part of a separate Byrka complex. The new radiometric age has been obtained at the Center of isotope analyses of the Karpinsky Russian Geological Research Institute (Saint-Petersburg) using a SHRIMP-II microprobe. Extracted zircons were colourless to yellow, transparent to cloudy ranging in size from 30 to 110 μm. Approximately 80 per cent of zircons were angular debris of elongated prismatic grains with a prominent magmatic zonation. The rest 20 per cent were subeuhedral grains. Eight zircon grains were used for geochronology analysis. The U–Pb isotope measurements were performed on five grain cores and three peripheral rims (Table 1).

Biostratigraphic markers for the Beletuy formation are limited and restricted to Ediacaran microphytolites and stromatolites (Stetsyuk 1977). Our field observations confirm this. The richest in stromatolites carbonate layers were found in the vicinity of the Byrka village. The stromatolites were identified as *Conophyton*, *Collumnaefacta*, *Colleniella*, *Gongylina* and *Stratifera*. The vast majority of microphytolites, identified as *Ambigolamellatus horridus Z. Zhur., Osagia minuta Z. Zhur., O. nimia Z. Zhur., O. donotella Korol., O. tenuilamellata Reitl., Tunicatella maritima Korol., Vesicularites compositus Z. Zhur.* and *Nubecularites catagraphus Reitl,* are common for the entire eastern Transbaikalia. The first three forms are typical for the Ediacaran (Kurilenko *et al.* 2002).

Available geological data and collected fauna fossils constrain the age of studied rocks between 600 and 535 Ma.

2.3 Zerentuy area

Both, Beletuy and comformably overlaying it Bystrin formations are well exposed in the Zerentuy area (Fig. 5). The most complete sections of the Bystrin formation are located in the vicinity of the Georgievka village. The formation is composed of interleaved throughout the section massive dolomites, calcareous dolomites with layers of sandstones, siltstones and siliceous sediments, which become more common in younger Cambrian deposits (Fig. 3). Samples from the Bystrin formation contain trilobites *Proerbia sp.*, various microphytolits: *Vesicularites, Nubecularites, Vermiculites* and *Osagia*, columnar and nodular–columnar stromatolites. Based on available biostratigraphic data (Kurilenko *et al.* 2002), the studied section have been deposited in the Early Cambrian between 529 and 517 Ma**.** Although sedimentary layers are well preserved and dip relatively shallow at some 30°–40° angle, field observations of biostratigraphic markers often reveal overturned bedding of the sedimentary strata. This observation indicates significant effect of regional tectonic processes on the studied sedimentary succession.

3 PALAEOMAGNETIC RESULTS

3.1 Sampling

Oriented palaeomagnetic core samples and fauna fossils were taken from the same outcrops. Palaeomagnetic sampling by default was accompanied by measurements of strata bedding elements. Overall, we have collected 148 palaeomagnetic samples from the Beletuy formation (56 samples from six sites), the Bystrin formation (78 samples from eight sites) and from Byrka dolerites (14 samples from two sites). Palaeomagnetic cores were oriented using a magnetic compass checked with sun compass readings.

The limestones of the Beletuy formation were sampled in Byrka area along the right flank of the Beletuy depression (localities 1 and 2 on Fig. 2). The same limestones were sampled

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Table 1. Zircon U–Pb dating results of Byrka dolerites.

Spot	$^{206}Pb_c$ (per cent)	U (ppm)	Th (ppm)	$^{232}Th/^{238}U$	$^{206}Pb^*$ (ppm)	(1) $^{206}Pb/^{238}U$ (age)	(2) $^{206}Pb/^{238}U$ (age)	(1) $^{207}Pb/^{206}Pb$ (age)	Discordant (per cent)	Total $^{238}U/^{206}Pb$	± Per cent	Total $^{207}Pb/^{206}Pb$	± Per cent	(1) $^{238}U/^{206}Pb^*$	± Per cent	(1) $^{207}Pb^*/^{206}Pb^*$	± Per cent	(1) $^{207}Pb^*/^{235}U$	± Per cent	(1) $^{206}Pb^*/^{238}U$	±Per cent	Error correction
215_1.1	0.54	362	134	0.38	12.9	260.8 ± 3.7	260.9 ± 3.8	256 ± 200	−2	24.09	1.4	0.0557	7	24.22	1.4	0.0513	8.8	0.292	8.9	0.04129	1.4	0.162
215_2.1	1.56	140	79	0.59	5.18	267.8 ± 4.8	267 ± 4.9	391 ± 350	46	23.2	1.7	0.0669	9.2	23.56	1.8	0.0545	15	0.319	16	0.04242	1.8	0.117
215_3.1	0.66	305	337	1.14	10.8	259.4 ± 3.8	259.1 ± 3.7	297 ± 130	15	24.19	1.5	0.0575	2.7	24.35	1.5	0.0523	5.9	0.296	6.1	0.04106	1.5	0.246
215_4.1	1.19	277	190	0.71	10.1	263.6 ± 4	262.8 ± 4.2	376 ± 270	43	23.66	1.5	0.0637	8.4	23.95	1.5	0.0541	12	0.312	12	0.04174	1.5	0.127
215_5.1	2.74	262	132	0.52	9.37	256.1 ± 3.9	256.3 ± 4.4	249 ± 350	−3	23.99	1.5	0.0731	9.2	24.66	1.6	0.0512	15	0.286	15	0.04052	16	0.101
215_6.1	1.11	322	353	1.13	11.6	261.6 ± 4	260.1 ± 4.5	471 ± 360	80	23.87	1.5	0.0653	12	24.14	1.6	0.0565	16	0.323	16	0.04142	1.6	0.097
215_7.1	1.5	363	385	1.1	12.8	254.9 ± 3.8	256.8 ± 3.8	−14 ± 280	−105	24.41	1.4	0.0579	5.8	24.78	1.5	0.0458	12	0.255	12	0.04034	1.5	0.129
215_8.1	1.61	363	167	0.71	8.63	256.9 ± 5	257.4 ± 4.1	204 ± 490	−21	24.19	1.5	0.0631	6.7	24.59	2	0.05	21	0.281	21	0.04066	2	0.092

Notes: Errors are 1σ; Pb_c and Pb^* indicate the common and radiogenic portions, respectively.
Error in standard calibration was 0.48 per cent (not included in above errors, but required when comparing data from different mounts).
(1) Common Pb corrected using measured ^{204}Pb.
(2) Common Pb corrected by assuming $^{206}Pb/^{238}U$–$^{207}Pb/^{235}U$ age concordance.

in Zerentuy area, southwest of the Gorny Zerentuy village (locality 3 on Fig. 5). Dolomites and sandstones of the Beletuy formation were sampled near the Ivanovka village (localities 4–6 on Fig. 5).

The limestones of Bystrin formation were sampled near the Byrka village along the both banks of the Verkhnyaya Borzya River (localities 7 and 8 on Fig. 2). At the Zerentuy area, one palaeomagnetic site was sampled on the left bank of the Urov River, while the main sampling area was located near the Georgievka village (localities 10–14 on Fig. 5).

Samples from two palaeomagnetic sites of Byrka dolerites were taken near the right flank of the Dalnyaya Byrka depression (localities 15 and 16 on Fig. 2).

3.2 Methods and equipment

Palaeomagnetic and rock-magnetic analyses have been conducted according to the standard techniques using instruments available in the Paleomagnetic Center of the Institute of Petroleum Geology and Geophysics SB RAS, Novosibirsk. Magnetic mineralogy was examined using an AGICO MFK1-FA magnetic susceptibility meter equipped with a high-temperature furnace. Temperature dependences of low-field magnetic susceptibility, $\kappa(T)$, were measured upon cycling from room temperature to 700 °C (in argon) with subsequent cooling to the room temperature. For the majority of measured powders, the $\kappa(T)$ curves were irreversible and revealed laboratory-induced alteration of magnetic minerals upon heating over 350 °C, resulting in neoformation of a magnetic phase with Curie temperatures typical for magnetite. According to this result, alternating field (AF) demagnetization was considered an optimal magnetic cleaning protocol for our samples. Magnetic remanence was measured with a 2G Enterprises 755 superconducting rock magnetometer equipped with an AF demagnetizing unit. After measurement of a natural remanent magnetization (NRM), AF stepwise demagnetization to the field up to 140 mT was performed on specimens prepared from each core sample. Directions of NRM components were determined using palaeomagnetic software with built-in analytical tools for principal component and great remagnetization circles analyses (Kirschvink 1980; McFadden & McElhinny 1988). Different modifications of the fold test were utilized in order to test the age of characteristic component of NRM (McFadden 1990; Watson & Enkin 1993; Enkin 2003).

3.3 Beletuy formation

NRM intensities and quality of palaeomagnetic signal are variable in samples from different rock units. The sandstones typically display relatively high NRM values in a range of 10^{-2}–10^{-1} A m^{-1}. Thermomagnetic curves for these samples appeared somewhat noisy. Characteristic Curie (Neel) temperatures for the main magnetic minerals were difficult to determine precisely. Heating cycles are characterized by a drop in magnetic susceptibility at 320 °C, indicative for the presence of pyrrhotite (Fig. 6a). Greater values of magnetic susceptibility during the cooling leg (on cooling from ∼500 °C) suggest formation of a new magnetic phase (probably magnetite) due to heating-induced transformation of pyrrhotite (e.g. Dekkers 1990) and/or reduction of haematite into a cubic ferromagnetic phase (similar to magnetite). The sandstones as well as dolomites revealed significantly non-uniform behaviour during AF demagnetization. The median destructive field (MDF) values vary from 5 to 40 mT. Palaeomagnetic signal is not stable,

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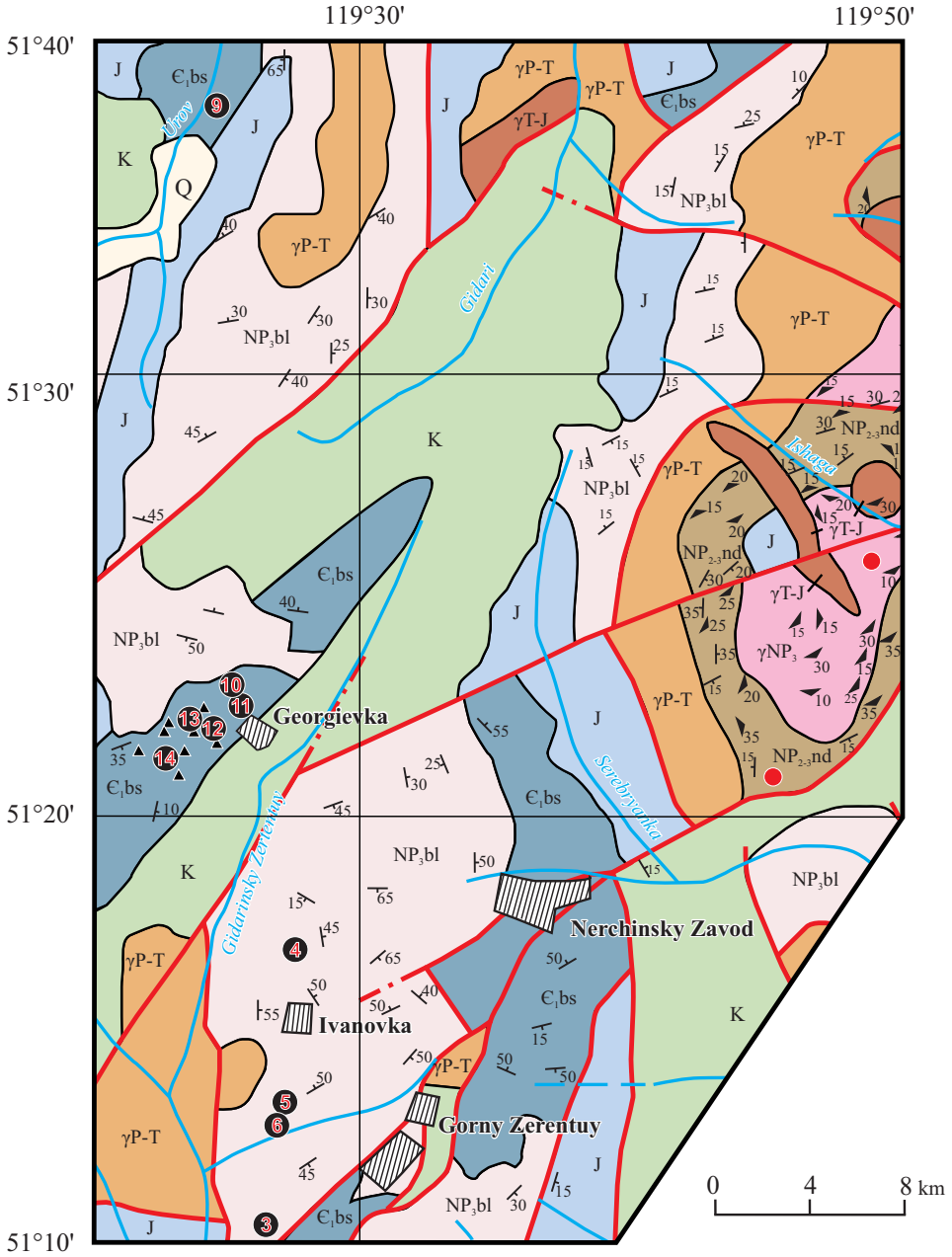


Figure 5. Geological map of Zerentuy area adopted from maps of Chita Geologic Survey (Chitageols'emka). Symbols are same with Fig. 2.

resulting in difficult to interpret vector endpoint diagrams. For the majority of our samples, NRM vectors tend to follow a great circle (Fig. 6b) on the equal area projections. Great circles intersect near the directions of the stable linear components, detected for some samples.

NRM of limestones was substantially weaker ($\sim 10^{-4}$–10^{-5} A m^{-1}). The $\kappa(T)$ dependences of limestones were also noisy (Fig. 7a). Heating curves, characterized by a gradual decrease in magnetic susceptibility at temperatures between 500 and 680 °C suggest the presence of magnetite and haematite. Cooling legs indicate significantly greater susceptibilities caused by formation of a new magnetic phase with Curie temperature of ∼580°C, typical for magnetite (Fig. 7a). The formation of the latter phase was likely due to the reduction of haematite during heating in argon.

Despite weak intensities of NRM in the limestones, palaeomagnetic signal in these samples is clearer and more stable than that in dolomites and sandstones.

The limestones from Zerentuy area have MDFs ∼ 20mT. Maximum demagnetization field of 140 mT destroys approximately 80 per cent of NRM (Fig. 7b), indicating low- to medium-coercivity magnetic minerals such as magnetite and titanomagnetite as the main carriers of natural remanence in these samples. In addition, a small amount of high-coercivity magnetic phase (haematite) is present in the samples. Nearly all measure samples manifested multiple-component NRM. The most stable component, isolated between ∼70 and 140 mT, was identified by straight linear demagnetization trajectory toward the origin on a vector endpoint diagram (Fig. 7b). This component was interpreted as characteristic remanence (ChRM).

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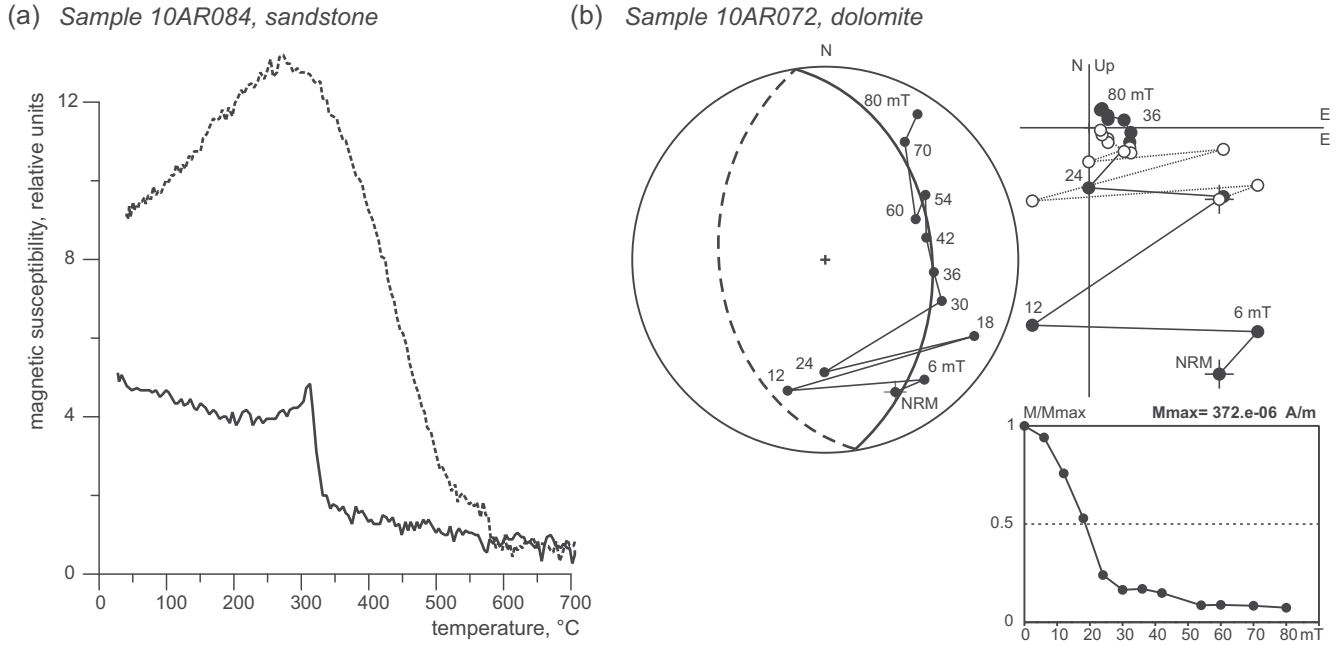


Figure 6. Examples of rock and palaeomagnetic results for Beletuy sandstones and dolomites: (a) typical dependence of low-field magnetic susceptibility (κ) versus temperature; solid (dashed) line shows temperature change during heating (cooling) cycle; (b) typical vector endpoint diagrams (open symbols are data points projections on vertical plane, solid symbols are projections on horizontal plane), equal area projection of NRM vector upon AF demagnetization and normalized NRM intensity plot.

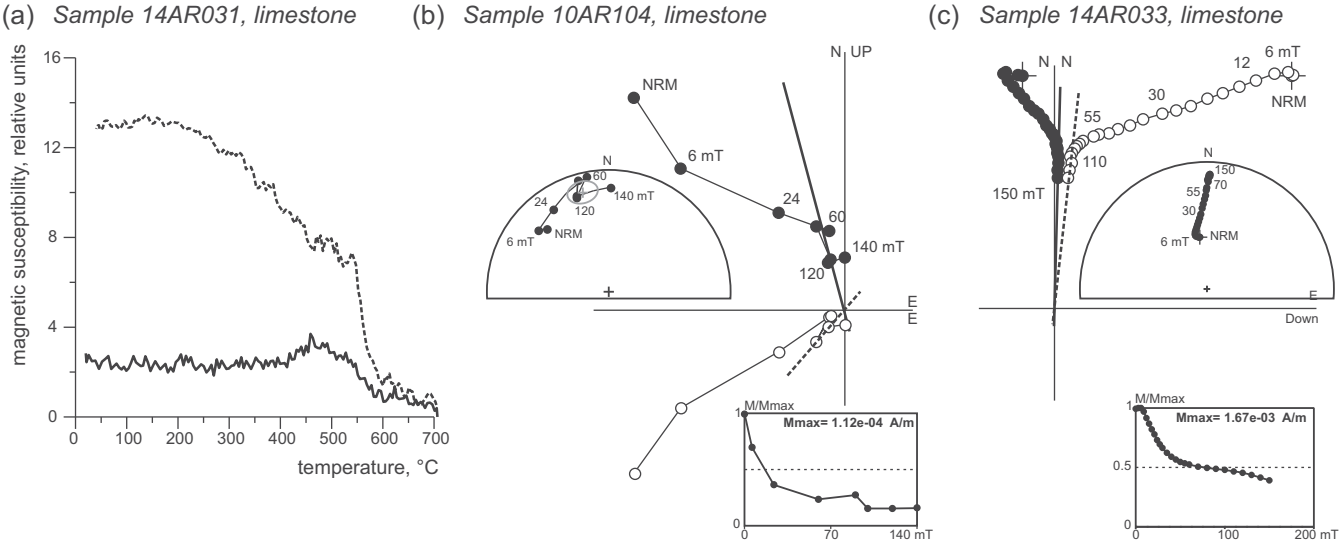


Figure 7. Examples of rock and palaeomagnetic results for Beletuy limestones: (a) typical dependence of low-field magnetic susceptibility (κ) versus temperature, solid (dashed) line shows temperature change during heating (cooling) cycle; (b) and (c) typical vector endpoint diagrams (open symbols are data points projections on vertical plane, solid symbols are projections on horizontal plane), equal area projection of NRM vector upon AF demagnetization and normalized NRM intensity plot.

The limestones from Byrka area are characterized by prevailing contribution of haematite in NRM. MDF values are ~70 mT. A 140 mT AF demagnetization step hardly removed ~60 per cent of the remanence. Nonetheless, the high-coercivity component of NRM linearly converges towards the origin of the vector endpoint diagrams (Fig. 7c).

The mean palaeomagnetic directions of the Beletuy formation have been scrutinized using different modifications of the fold test in order to test their relative age. The parametric test of Watson & Enkin (1993) gave the best clustering of the directions upon 100.8 ± 5.2 per cent untilting (Fig. 8a). The correlation fold test (Enkin 2003) was also positive (DC slope: 0.97 ± 0.09). We also used the McFadden (1990) fold test, in which a calculated statistic ξ is compared to the tabulated critical value for the total number of sites on both limbs (N). If ξ exceeds the critical value, then the null hypothesis that the palaeomagnetic directions do not correlate with tectonic in unfolded position is rejected. Results of the correlation test: *in situ* $\xi_1 = 4.093$, tilt corrected $\xi_1 = 0.001$ and $\xi_{\text{critical}} = 2.86$, further attest the pre-folding age of the ChRM in studied rocks at 95 per cent confidence.

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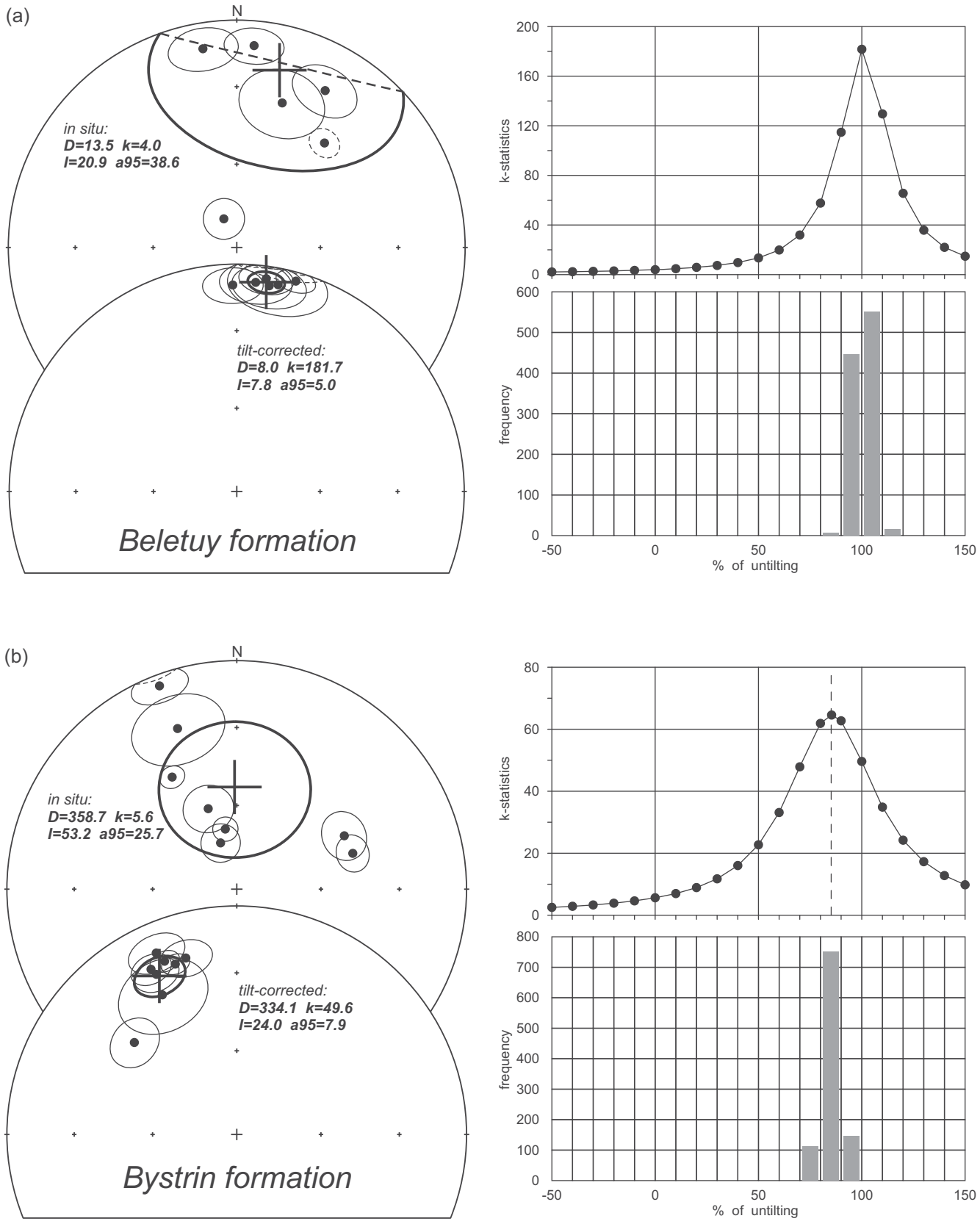


Figure 8. Palaeomagnetic directions with the 95 per cent confidence circle before (upper plot) and after (lower plot) tilt correction and corresponding diagrams showing results of the fold test of Watson & Enkin (1993).

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We think that the calculated mean palaeomagnetic direction faithfully describes the geomagnetic field at the time of Beletuy sedimentation. Although it is possible that part of natural remanence is not detrital, but rather chemical, presence of magnetite and positive fold tests indicate that characteristic component of NRM is likely primary and corresponds to time of deposition. Considering age, total thickness and significant duration of deposition of studied rocks, we believe that calculated palaeomagnetic pole (Table 2) truly represents the time-averaged geomagnetic field. The palaeomagnetic pole can be assigned the age of ∼560 Ma.

3.4 Bystrin formation

Rock-magnetic characteristics of the Bystrin limestones are similar to those sampled from the Beletuy succession. NRM intensities are relatively low and vary in a range between 10^{-4} and 10^{-5} A m^{-1}. Samples yielded irreversible $\kappa(T)$ curves that have a bump at ∼250–400 °C (Fig. 9a) and gradual decrease in magnetic susceptibility at higher temperatures with an inflection close to the Curie temperature of magnetite. About 30 per cent higher susceptibilities are typical for cooling curves. The latter indicates laboratory-triggered alteration of samples with formation of magnetite. A small decrease in magnetic susceptibility during heating above 600 °C suggests the presence of some amount of haematite.

MDF values varied between 15 and 60 mT. Almost complete demagnetization was usually achieved at 70 mT. Sometimes a small part of remanence remained stable throughout an entire demagnetization sequence up to 140 mT. This observation further confirms low to intermediate-coercivity magnetic minerals as the main remanence carriers with a small contribution from the high-coercivity magnetic phase. Vector endpoint diagrams often reveal a single-component NRM with the direction converging to the origin. The ChRM directions are well defined (Figs 9b and c).

All sampled limestones of the Bystrin formation (except for two samples from site 10AR01, e.g. Fig. 9b) have normal polarity of palaeomagnetic directions. This result differs significantly from the previous study (Bretstein & Klimova 2007).

Although the result of McFadden (1990) correlation fold test is inconclusive, Enkin (2003) test gave positive result (DC slope: 0.84 ± 0.19). A maximum *k*-statistics obtained using the Watson & Enkin (1993) test has been achieved at 85.5 ± 7.0 per cent of unfolding (Fig. 8b). Therefore, we conclude that magnetization in sampled rocks is either pre- or sin-folding, acquired before or during the early stage of regional deformation. Taking into account the rock-magnetic data and age of the studied sequences, we think that isolated ChRM component is very likely primary. Sampled rocks represent a time interval significant enough to record secular variations, making the palaeomagnetic pole, calculated from individual virtual geomagnetic poles, a reliable time-averaged reading of the geomagnetic field. New palaeomagnetic pole has been assigned the age of ∼525 Ma.

3.5 Byrka dolerite

The Byrka dolerite has average NRM values at about 10^{-3} A m^{-1}. AF demagnetization showed that the MDF varies from 15 to 20 mT. In all samples, beginning from 10 to 12 mT until full demagnetization in field of 80–100 mT, a ChRM direction can be confidently detected (Fig. 10). Nonetheless, the ChRM directions of some individual samples differ significantly from each other within sampling sites (Fig. 10), and do not form compact clusters. Thus, these rocks apparently underwent remagnetization that destroyed the initial palaeomagnetic information, and are therefore unfit for analysis.

4 DISCUSSION

In its traditional palaeogeographic reconstructions, Argun terrane is considered a fragment of Amuria—a microcontinent, which core is built of Palaeoproterozoic crystalline complexes, overlain by Neoproterozoic to Cambrian sedimentary sequences. These sequences include the two studied Beletuy and Bystryn formations. However, recent isotope-geochemistry and geochronology data indicate significantly younger age of the metamorphic basement (Golubev *et al.* 2010). Our Sm−Nd analysis of metamorphic rocks and granites from the eastern part of the Zerentuy area (known as Ishagin granitic–gneiss dome) confirms the results of Golubev *et al.* (2010). T_{Nd}(DM) data for metamorphic rocks yielded a protolith age of 1313 Ma with negative ε_{Nd} value of ∼ –8.7. Similar T_{Nd}(DM) age of 1247 Ma and ε_{Nd} about –5.7 were obtained from intruded granites. Consistent isotope data reported in two independent studies do not support Palaeoproterozoic age of the Zerentuy area basement. We think that granitic–gneiss systems exposed at the territory of Argun terrane are associated with accretion-collisional processes that took place during the early Ediacaran stage of the Mongol–Okhotsk Ocean evolution. These structures are manifested along the entire southeastern margin of Siberian craton (Gordienko & Metelkin 2016). For example, Ediacaran (583 ± 6 and 607 ± 8 Ma) island arc type gabbro-diorite-granite series have been recognized within the Mamyn metamorphic protrusion, located to the east from the studied area (Sorokin *et al.* 2015). Thus, in the base of large crystalline structures, formerly interpreted as microcontinents, no typical Archean–Palaeoproterozoic rocks have been recognized. Accordingly, Argun terrane cannot be considered a microcontinent involved in formation of a rigid carcass of Amuria. This interpretation is further corroborated in studies of potential candidates for cratonic fragments of Amuria, such as Mamyn, Khangay and other terranes. In most cases, the oldest parts of these terranes are made of reworked oceanic island arc-related crust, intruded by granites no older than 880 Ma (Yarmolyuk *et al.* 2015). These observations cast a doubt on the formation of the tectonic blocks due to breaking of ancient continental crust after the Rodinia breakup (Gordienko 1994; Dobretsov *et al.* 2013). But instead indicate accretional growth of the crust on the Siberian margin during the Ediacaran stage of the Mongol–Okhotsk Ocean as a part of the Palaeo-Asian Ocean evolution (Metelkin 2013; Gordienko & Metelkin 2016). Unfortunately, palaeomagnetic record for island arc terranes that could have been involved in these processes is extremely poor. Therefore, we lack evidence that accretion basement of Argun terrane has formed directly on the margin of Siberia. However, this scenario is possible to test using data from the Beletuy and Bystryn formations.

Firstly, fauna fossils identified in rocks of the Beletuy and Bystrin formations are common for the entire MOFB and also similar to those from Siberian sedimentary basins (Popeko *et al.* 1993; Parfenov *et al.* 1999). This observation can be explained by a development of a wide Ediacaran–Cambrian sea or series of connected basins, formed on the accreted margin of Siberian craton.

Secondly, new palaeomagnetic poles are similar to the coeval poles for Siberia (Table 2 and Fig. 11). Moreover, the new palaeomagnetic pole for the Beletuy formation does not differ significantly

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Table 2. Summary of achieved and expected palaeomagnetic data for Argun terrane.

Number	Site, rock type	Site coordinates		*n*/*N*	*In situ*		Tilt corrected		*k*	α_{95}	Pole			PL
		GLat	GLong		*D* (°)	*I* (°)	D (°)	I (°)			PLat	PLong	dp dm^{-1} or A_{95}	
	Beletuy formation													
1	14AR02, limestone	50.6568	118.6627	7/10	335.9	78.8	7.9	6.2	73.4	7.3	−41.7	108.4	3.7/7.3	− 3.1 ± 5.2
2	14AR04, limestone	50.6939	118.6561	10/11	4.9	11.9	358.9	10.1	44.2	7.4	−44.1	120.5	3.8/7.5	− 5.1 ± 5.3
3	10AR11, limestone	51.1951	119.4413	9/10	17.4	33.8	11.3	8.1	15.1	13.7	−42.1	103.7	7.0/13.8	− 4.1 ± 9.8
4	10AR08, dolomite	51.2882	119.4678	5^{2}/6	350.4	12.4	5.2	8.5	61.9	9.8	−43.1	111.9	5.0/9.9	− 4.3 ± 7.0
5	10AR09, sandstone	51.2190	119.4617	9^{8}/10	29.3	22.0	9.0	9.3	26.4	10.2	−43.1	106.7	5.2/10.3	− 4.7 ± 7.3
6	10AR10, sandstone	51.2166	119.4600	9^{7}/9	39.9	−39.9	15.7	4.4	95.3	5.3	−39.4	98.5	2.7/5.3	− 2.2 ± 3.8
	Mean				13.5	20.9			4.0	38.6				
							8.0	7.8	181.7	5.0	−42.4	108.1	4.8	− 3.9 ± 3.6
	Expected, Argun terrane (Bretstein & Klimova 2007)						196.6	−8.4		35.8	−33.0	99.2	25.6	− 4.2 ± 25.6
	Expected, Siberian craton (Metelkin *et al.* 2012)						237.8	−24.6		9.4	−30.0	46.7	7.4	− 12.9 ± 7.4
	Bystrin formation													
7	14AR01, limestones	50.6397	118.6873	8/10	330.2	42.9	340.3	21.9	138.0	4.2	−47.2	148.1	2.3/4.4	− 11.4 ± 3.2
8	14AR03, limestones	50.6497	118.5951	10/10	340.6	59.4	312.1	39.8	28.8	8.7	−43.5	189.8	6.3/10.5	− 22.6 ± 8.1
9	10AR01, limestones	51.5848	119.4137	10/10	339.3	5.5	344.0	20.8	39.3	7.8	−47.7	142.7	4.3/8.2	− 10.8 ± 5.9
10	10AR03, limestones	51.3802	119.4407	10/10	349.3	68.2	337.5	19.1	121.5	4.4	−44.9	151.1	2.4/4.6	− 9.8 ± 3.3
11	10AR07, limestones	51.3799	119.4405	8/9	340.8	72.6	333.5	22.8	65.4	6.9	−45.3	157.4	3.9/7.3	− 11.9 ± 5.3
12	10AR04, limestones	51.3747	119.3919	10/10	72.8	46.1	332.7	19.8	61.7	6.2	−43.5	157.5	3.4/6.5	− 10.2 ± 4.7
13	10AR05, limestones	51.3755	119.3916	6/9	339.3	26.1	332.1	31.5	22.3	14.5	−49.4	162.4	9.1/16.3	− 17.0 ± 12.2
14	10AR06, limestones	51.3566	119.3814	9/10	63.5	46.6	336.2	14.3	39.4	8.3	−42.0	151.6	4.3/8.5	− 7.3 ± 6.1
	Mean				358.7	53.2			5.6	25.7				
							334.1	24.0	49.6	7.9	−46.2	157.5	7.1	− 12.6 ± 6.2
	Expected, Argun terrane (Bretstein & Klimova 2007)						187.0	−8.9		8.1	−43.1	109.4	5.8	− 4.5 ± 5.8
	Expected, Siberian craton (Cocks & Torsvik 2007)						162.4	−14.3		30.0	−43.9	143.6	22.0	− 7.3 ± 22.0
	Byrka dolerites													
15	14AR05, dolerite	50.5398	118.8553	5/6	204.7	−59.3	193.9	3.7	1.1	180				
16	14AR06, dolerite	50.5382	118.8349	8/8	229.5	−3.6	231.2	−3.4	2.4	46.2				

Notes: Numbers in the first column correspond to sampling sites on Figs 2, 5, 8 and 11; *n*/*N*—number of samples used in statistics to all studied samples (superscript denotes number of remagnetization circles); *D*—declination; *I*—inclination; *k*—Fisher (1953) precision parameter; α_{95}—radius of the 95 per cent confidence circle; palaeomagnetic pole was calculated assuming reversed polarity of geomagnetic field; dp dm^{-1}—semiaxes of 95 per cent confidence ellipse; mean (palaeomagnetic) pole was calculated as the mean of virtual geomagnetic poles, where A_{95}—radius of the 95 per cent confidence circle and PL—calculated palaeolatitude for Argun terrane; expected palaeomagnetic directions were calculated using an averaged location at 51.0°N. 119.0°E.

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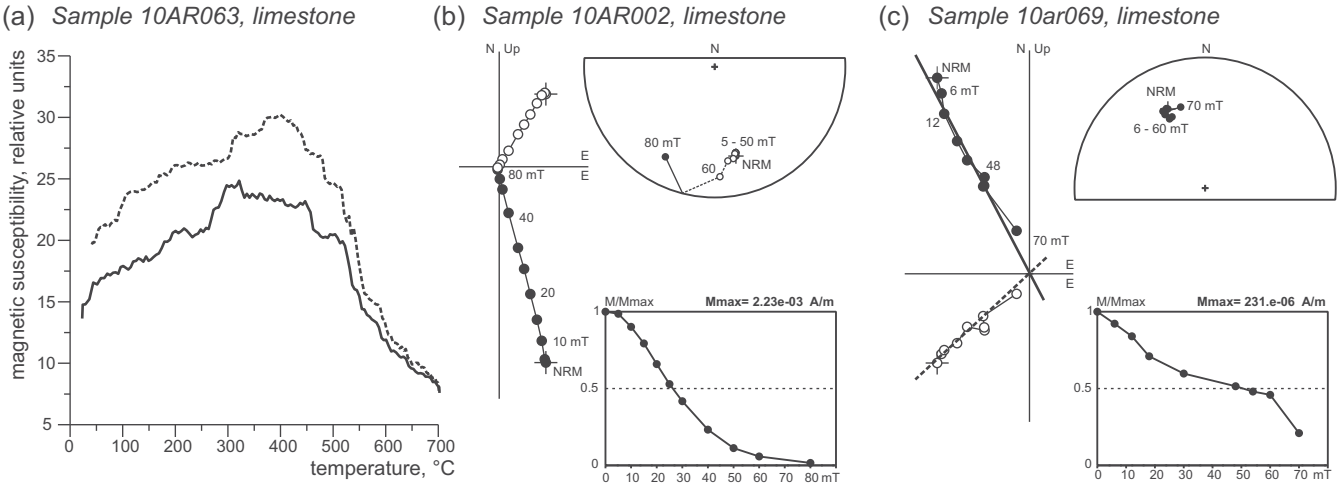


Figure 9. Examples of rock and palaeomagnetic results for Bystrin limestones: (a) typical dependence of low-field magnetic susceptibility (κ) versus temperature, solid (dashed) line shows temperature change during heating (cooling) cycle; (b) and (c) typical vector endpoint diagrams (open symbols are data points projections on vertical plane, solid symbols are projections on horizontal plane), equal area projection of NRM vector upon AF demagnetization and normalized NRM intensity plot.

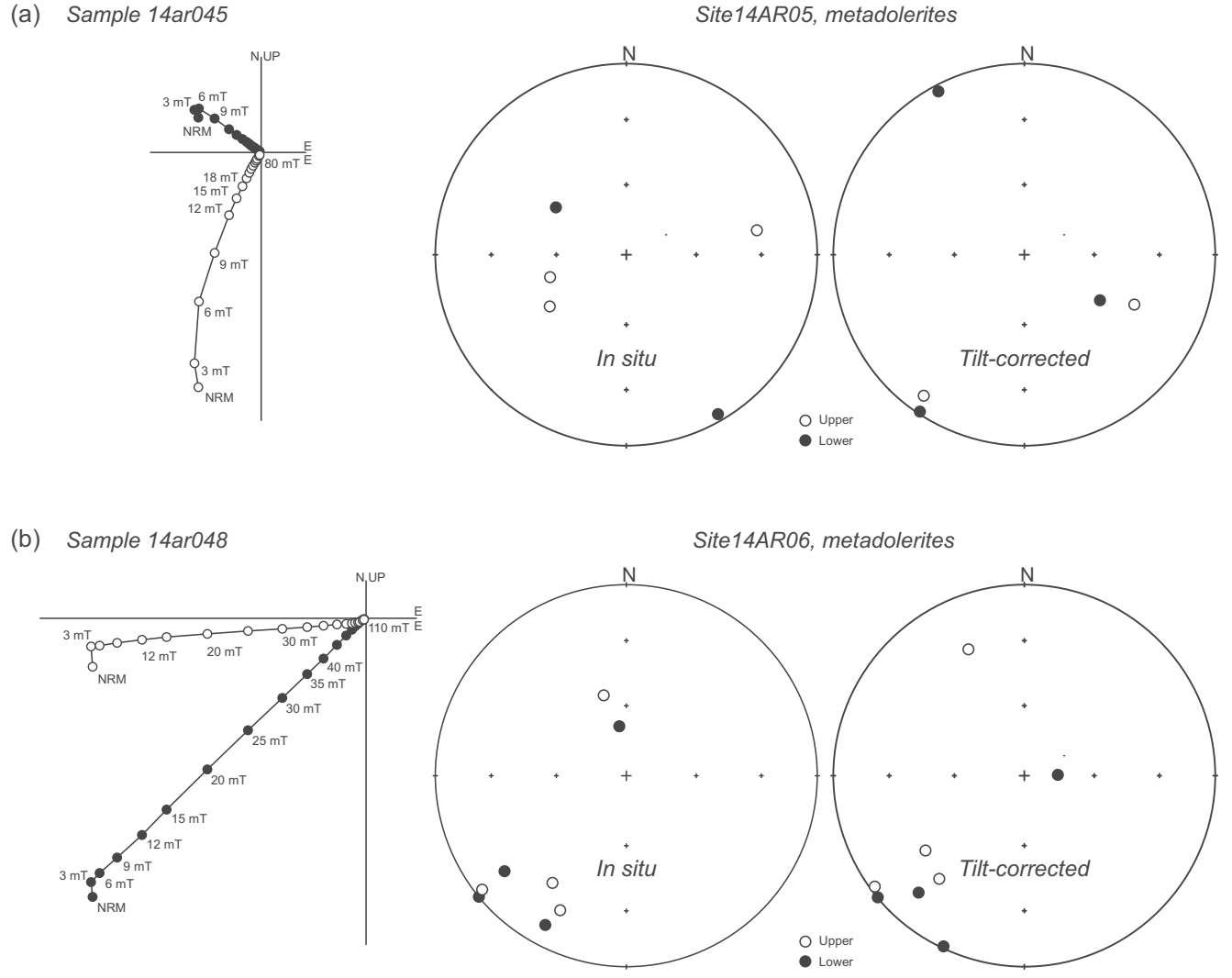


Figure 10. Examples of palaeomagnetic results for Byrka dolerites: (a) from site 14AR05 and (b) from site 14AR06.

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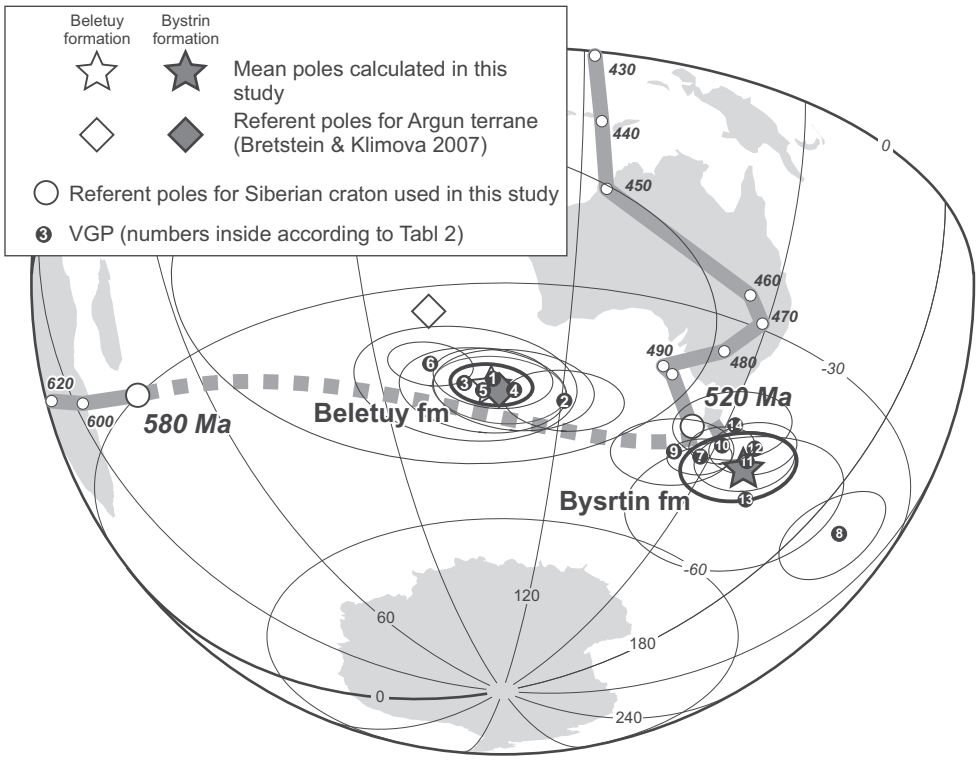


Figure 11. Palaeomagnetic poles calculated for Argun terrane. Solid line shows apparent polar wander (APWP) for Siberia. Dashed line is a hypothetical APW path for the ~520–580 Ma period. Palaeomagnetic data were adopted from Metelkin *et al.* (2012) and Cocks & Torsvik (2007). Numbers next to palaeomagnetic poles indicate their absolute ages.

from that of the previous study (Bretstein & Klimova 2007). Although Ediacaran segment of apparent polar wander path (APWP) for Siberia is widely debated, most of the available palaeomagnetic data [see summary in Metelkin *et al.* (2012)] are in good agreement with our results (Fig. 11). Therefore, we propose that Argun terrane had relatively similar to the present-day position with respect to Siberia already in Ediacaran.

Based on palaeomagnetic data obtained from the Bystrin formation, relative configuration of Argun terrane and Siberia remained the same in early Cambrian. Considering a predominant reversed polarity of geomagnetic field between 529 and 517 Ma, the Bystrin formation palaeomagnetic pole plots close to the coeval pole for Siberia (Fig. 11). The new pole position is ~45° away from the previously reported by Bretstein & Klimova (2007). However, as pointed out by the authors, this pole is very likely biased by secondary components of NRM (Bretstein & Klimova 2007). Absence of strong secondary remanence components and positive fold test are strong indicators that our palaeomagnetic data were less affected by secondary overprints and represents ChRM with age close to that of sedimentation.

Available biostratigraphic and palaeomagnetic data indicate that the Beletuy and Bystrin basins were located at equatorial latitudes (palaeolatitudes vary from 3.9° ± 3.6° to 12.6° ± 6.2° S) close to Siberian margin (see on Table 2).

On the other hand, our data are in conflict with some geological conclusions about Mongol–Okhotsk Ocean between Argun terrane and Siberian craton during the Ordovician (Bulgatov *et al.* 2010; Gordienko *et al.* 2012). A narrow strip of presumably Ordovican oceanic complexes located at the northern part of Argun terrane is included in Mongol–Okhotsk suture (Fig. 1). According to palaeomagnetic data from Oldoy terrane, situated to the east of our studied area, this oceanic basin width could have reached 3500 km by the Devonian (Kravchinsky & Sorokin 2001). However, recently Sorokin *et al.* (2015) reported new data that indicate island arc rather than oceanic origin of Oldoy terrane. Therefore, relation of Oldoy terrane to an Ordovican ocean, separating Argun terrane and Siberia is questionable. Nonetheless, this does not explain controversy in palaeomagnetic data that suggest a rather complex tectonic evolution of Argun terrane. In this somewhat non-realistic scenario, Argun terrane could have been connected to Siberian continental margin already in the Neoproterozoic. Rifted away from Siberian craton in Ordovician, the terrane experienced an independent (or within Amuria) history during the late Palaeozoic, with subsequent collision with the same part of Siberia in the Mesozoic during closure of the ocean, retaining exactly the same relative orientation.

Here, we propose an alternative, more simple model in which Ordovician oceanic rocks between Argun terrane and Siberian craton indicate a narrow spreading basin (similar to that of the Red Sea) that was formed during Palaeozoic stage of the Mongol–Okhotsk Ocean evolution. The spreading basin was too narrow to be unambiguously detectable using palaeomagnetic data. Argun terrane was involved in Palaeozoic–Mesozoic evolution of the MOFB, but instead was located near the Siberian margin and, therefore, was not a part of Amuria, if the latter existed on the other side of the ocean, as proposed in conventional reconstructions (Zonenshain *et al.* 1990; Kravchinsky *et al.* 2002; Parfenov *et al.* 2003; Golonka *et al.* 2006).

5 CONCLUSIONS

(1) Strong similarities between Argun terrane and adjacent Neoproterozoic–Palaeozoic structures of the MOFB point at their close association. The similarities are manifested in composition, lack of Archean–Palaeoproterozoic structures, and mainly early

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Neoproterozoic age of the oldest igneous and metamorphic rocks protoliths. Argun terrane must be considered an accretion-collision structure amalgamated to the Siberian craton mainly during the Neoproterozoic.

(2) Ediacaran–Cambrian fauna fossils found in sediments of Argun terrane are similar to those described for Siberia and adjacent sedimentary basins of the MOFB. This indicates biofacial unity of these areas, same fauna migration ways, and therefore we recognize the existence a wide continental margin basin during this time.

(3) Palaeomagnetic data suggest equatorial position of Argun sedimentary basin between 560 and 525 Ma. Palaeomagnetic poles are virtually the same with the coeval poles for Siberia, further attesting spatial and temporal association of Argun terrane and Siberian craton. We think that Argun terrane was a part of Siberia since the Ediacaran and was never associated with the North China Craton.

(4) Argun terrane was not involved in formation of the composite structure of Amuria, located in classical models on the opposite margin of Mongol–Okhotsk Ocean. Furthermore, our data point at insufficient evidence for allotment of Amuria as a separate superterrane in principal. Although our data call into question conventional palaeogeographic schemes, we realize that additional high-quality palaeomagnetic data for the Ediacaran are needed to resolve ambiguities of palaeogeographic reconstruction for this period of Earth history.

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